

8/10

64 byte $\rightarrow$
1 bit

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ECE270: ELD: Quiz 3A (20 Minutes)

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1. Write a Verilog code for memory of size 64 bytes with each location containing 1 byte. The read and write operation must be synchronous. The write operation is controlled via write enable (WEN) signal. The memory should allow one independent write and two independent read operations simultaneously. (8 Marks)
2. Explain the difference between three AXI protocols. (2 marks)

1. module (~~input wen~~,

input wen,

input clk,

input ~~waddr~~ [5:0] waddr,

input [5:0] raddr1,

input [5:0] raddr2,

input [7:0] din,

output [7:0] out1,

output [7:0] out2);

reg [7:0] mem [5:0] = 0;

reg [7:0] rout1 = 0;

reg [7:0] rout2 = 0;

always @ (posedge clk) begin

if (wen)

mem[waddr] <= din;

rout1 <= mem[raddr1];

rout2 <= mem[raddr2];

else

rout1 <= mem[raddr1];

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route2 <= mem [addr2];

end

assign out1 = route1;

assign out2 = route2;

endmodule

2. Three types of AXI protocols are AXI memory map, AXI lite and ~~AXI~~ AXI stream.

AXI memory: has 5 channels, ~~Read~~ Read data, Read address and control, write data, write address and control and write response

~~The data~~ The ^{no. of} ~~size~~ of the burst can be ~~upto~~ more than 1. For bulky data.
per transaction

AXI lite: has same 5 channels as the memory map but the ^{no. of} burst ~~size~~ is only 1. For small data.
per transaction

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AXI stream: has only write data channel with capability for bulky data

- Source of info is slave in Read data & write response channel if channel exists in protocol
- all channels in all protocols have minimum data, valid and ready signal
- all data channels also have a last signal
- all protocols have master and slave side, where master initiates transaction, slave responds.
- Source of information is master in Read Address and control, write data, write address and control channels ~~in all~~ ~~the~~ protocols. if channel exists in protocol.